

Harmain Ali Butt

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PROFESSIONAL SUMMARY

Highly accomplished Lead Software Engineer with 8+ years of experience in architecting, developing, and deploying robust software solutions, specializing in Artificial Intelligence (AI) and Machine Learning (ML) models within production environments. Proven expertise in full-stack development, including Python, C++, and JavaScript, alongside advanced frameworks like PyTorch and TensorFlow. Adept at designing scalable ML/AI tools, implementing rigorous quality control protocols, and automating complex data analytics pipelines. Experienced in leveraging web development technologies such as Flask and Angular to create interactive applications and committed to ensuring high performance and reliability of deployed systems.

TECHNICAL SKILLS

Python, C++, JavaScript, Docker, AWS, Git, SQL, PostgreSQL, MongoDB (noSQL), RESTful APIs, Flask, Angular, ASP.Net Core, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Artificial Intelligence (AI), Machine Learning (ML), Data Science, Algorithms, Model Deployment, Quality Control, Data Analytics Automation, Linear Algebra, Statistics, Jupyter Notebooks, VS Code, Google Colab

PROFESSIONAL EXPERIENCE

Senior Machine Learning Engineer — AMD (Jan 2024 - Present)

Technologies: Python, PyTorch, TensorFlow, C++, AWS, Docker, SQL, noSQL

Led the development and deployment of advanced Artificial Intelligence (AI) models for optimizing GPU performance, resulting in a 15% improvement in processing efficiency across key product lines. Implemented robust quality control protocols and established automated monitoring systems to meticulously track output data quality in production environments, reducing critical errors by 20%. Architected and developed scalable ML/AI tool development pipelines using Python and C++, integrating complex algorithms and linear algebra concepts to enhance model training and inference capabilities. Collaborated with cross-functional teams to automate data analytics processes, identifying critical opportunities for ML and AI applications that drove strategic product enhancements.

Machine Learning Engineer — NVIDIA (Jan 2021 - Dec 2023)

Technologies: Python, TensorFlow, NumPy, Pandas, JavaScript, Flask, PostgreSQL

Developed and deployed machine learning models for real-time image recognition and natural language processing, significantly improving data processing speeds by 25% using TensorFlow and Python. Contributed to ML/AI tool development, creating custom libraries that integrated advanced statistics and algorithms for feature engineering and model evaluation. Designed and implemented interactive web development interfaces for ML model visualization and interaction using JavaScript and Flask, enhancing user accessibility and data interpretation. Established rigorous quality control measures for deployed models, including continuous monitoring of output data quality in production environments to ensure model integrity and performance.

AI Developer — Google (Jan 2018 - Dec 2020)

Technologies: Python, Scikit-learn, C++, SQL, Angular, ASP.Net Core

Developed core Artificial Intelligence (AI) components for Google Cloud services, focusing on predictive analytics and anomaly detection using Python and C++. Engineered and deployed machine learning models, applying principles of linear algebra and statistics to optimize data-driven decision-making processes. Collaborated on web development projects, building user interfaces with Angular and ASP.Net Core to expose ML model functionalities and insights to internal stakeholders. Participated in the design of data pipelines for automating data analytics, ensuring data quality and consistency for downstream ML applications.

EDUCATION

Master's in Computer Science, specializing in Artificial Intelligence — Stanford University (2017)

Bachelor of Science in Computer Engineering — University of California, Berkeley (2016)

CERTIFICATIONS

Machine Learning Engineering with TensorFlow on Google Cloud | Google Cloud, Deep Learning Specialization | DeepLearning.AI (Andrew Ng), Professional Certificate in Python for Data Science and AI | IBM, Advanced JavaScript and Front-End Development | Udemy